

Phase 7 / Step 7.12: Molecular baseline (H2, NIST WebBook diatomic constants)

State (ground)	X 1Σg 1σ2
T_e	0 cm ⁻¹
ω_e	4401.213000 cm ⁻¹ (≈0.545681 eV)
ω_e x_e	121.336000 cm ⁻¹ (≈0.015044 eV)
B_e	60.8530000 cm ⁻¹ (≈0.00754481 eV)
α_e	3.0622000 cm ⁻¹ (≈0.00037966 eV)
D_e (distortion)	0.0471000 cm ⁻¹ (≈0.0000058397 eV)
r_e	0.741440 Å (=7.414400e-11 m)
Morse (derived)	D ₀ ≈ 4.6793 eV, D _e ≈ 4.6793 eV (from ω _e , ω _e x _e)

Data: NIST Chemistry WebBook (Huber & Herzberg compilation)
This figure fixes baseline targets for Part III; it is not a P-model derivation.